

Optimal Bandwidth for Multitaper Spectrum Estimation

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Abstract—A systematic method for bandwidth parameter selection is desired for Thomson multitaper spectrum estimation. We give a method for determining the optimal bandwidth based on a mean squared error (MSE) criterion. When the true spectrum has a second-order Taylor series expansion, one can express quadratic local bias as a function of the curvature of the spectrum, that can be estimated by using a simple spline approximation. This is combined with a variance estimate, obtained by jackknifing over individual spectrum estimates, to produce an estimated MSE for the log spectrum estimate for each choice of time-bandwidth product. The bandwidth that minimizes the estimated MSE then gives the desired spectrum estimate. Additionally, the bandwidth obtained by using our method is also optimal for cepstrum estimates. We give an example of a damped oscillatory (Lorentzian) process in which the approximate optimal bandwidth can be written as a function of the damping parameter. The true optimal bandwidth agrees well with that given by minimizing estimated the MSE in these examples.

Index Terms—Bandwidth, cepstrum, multitaper, spectrum analysis.

I. INTRODUCTION

NONPARAMETRIC spectrum estimation is an indispensable tool in the exploratory data analysis of univariate time series. No unbiased spectral estimator exists for a discrete, finite time series, however, *bandwidth* is a quantity through which the statistical properties of a nonparametric spectrum estimator can be understood. That is, bandwidth controls the natural tradeoff between bias and variance of the estimator. We are concerned with the bandwidth optimality in nonparametric spectrum estimation, specifically for multitaper estimates [1], [2] that require a user-defined bandwidth.

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The multitaper method offers considerable performance advantages over nonparametric single-taper [3, Fig. 2] and parametric estimates [4, Figs. 3, 6, and 9] [5]. Stoica and Sundin [6] showed that multitaper estimators are maximum likelihood when the spectrum is the sum of step functions with “treads” no smaller than the chosen bandwidth. Bronez [7] showed that when two of bias, variance, and bandwidth are fixed, the multitaper method produces dramatically smaller values for the third quantity than does the Welch method [8], which generalizes the periodogram by averaging spectrum estimates computed on overlapped segments of data. Prieto *et al.* [9] gave a procedure for eliminating a local quadratic bias term in the multitaper estimate; see also Riedel and Siderenko [10]. No systematic method has been described for bandwidth choice. Most computer implementations use a default time-bandwidth product fixed at 4, regardless of the context in which the spectrum estimation is being applied.

The main contribution of this letter, beginning in Section VI, is a new methodology to estimate optimal bandwidth by minimizing the estimated mean squared error (MSE) of the log spectrum as a function of time-bandwidth parameter. The MSE is obtained from the data by separately estimating bias, Section IV, and variance, Section V. Using a second-order Taylor expansion for the spectrum as in [1] and [9] bias is described by the second derivative of the spectrum. In Section IV-B, the second derivative is estimated by using splines. Variance estimation is done in Section V-A by using the jackknife technique described in [11]. In Section VII an illustrative example is given where the optimal bandwidth is known; in this case, minimization of estimated MSE produces bandwidth close to the optimal value. This method greatly reduces the subjectivity in bandwidth choice and is easily automated for simple implementation. We note that optimal bandwidth for the log-spectrum is also optimal for the multitaper cepstrum [12].

II. MULTITAPER SPECTRUM ESTIMATION

Let $x(t)$, for $t = 0, 1, \dots, N - 1$, be a zero-mean stationary time series with unit sampling and spectral density $S(f)$ [13]. Consider the simple *multitaper* estimate of the spectrum, $\bar{S}(f)$, computed by

$$\widehat{S^{(k)}}(f) = \left| \sum_{t=0}^{N-1} v_t^{(k)} x_t e^{-i2\pi ft} \right|^2, \quad \bar{S}(f) = \frac{1}{K} \sum_{k=0}^{K-1} \widehat{S^{(k)}}(f) \quad (1)$$

which is an average of K approximately independent estimates of the spectrum. The averaging of independent estimators decreases the variance of the overall estimate. The *discrete prolate spheroidal sequences*, or Slepian sequences $v_t^{(k)}(N, W)$, where

W is the bandwidth [14], solve the eigenvalue problem

$$\lambda_k(N, W)v_s^{(k)}(N, W) = \sum_{t=0}^{N-1} \frac{\sin 2\pi W(t-s)}{\pi(t-s)} v_t^{(k)}(N, W)$$

for $s = 0, \dots, N-1$. For ease of notation, $v_t^{(k)}(N, W)$ and $\lambda_k(N, W)$ will be written $v_t^{(k)}$ and λ_k , respectively. The eigenvalues (concentrations) in decreasing order $1 > \lambda_0 > \lambda_1 > \dots > \lambda_{N-1} > 0$ have only $K \approx 2NW$ large values close to one, after which they drop sharply to zero. The Slepian functions form an orthonormal set in time; and the *Slepian functions*,

$$U^{(k)}(f) = \sum_{t=0}^{N-1} v_t^{(k)} e^{-i2\pi ft} \quad (2)$$

are orthonormal on $(-1/2, 1/2)$ and orthogonal on $(-W, W)$. This orthogonality property ensures the approximate independence of the quantities $\hat{S}^{(k)}(f)$ in (1), thus controlling the variance of $\bar{S}(f)$. The approximate bandlimitedness of the tapers controls the bias.

III. BANDWIDTH CHOICE

As mentioned, the bandwidth controls the bias-variance trade-off in spectrum estimation, so bandwidth choice can be based on either the desired resolution or desired statistical properties. In the former case, if one wishes to resolve two peaks spaced approximately $2W_0$ Hz apart, one should ensure that W is less than W_0 . In the latter case, larger bandwidth can generally be used with larger sample size (to favor variance over bias). But in this case, many authors prefer to control variance by using a multitaper-Welch method [3], [8].

Most often, bandwidth choice is often carried out by simply using $NW = 4$. The reasons for this choice may be the following:

- 1) Thomson uses this choice in the original letter [1] for a short series ($N = 100$);
- 2) larger NW results in additional computation time;
- 3) canned multitaper estimators in standard software packages generally use $NW = 4$ by default; and
- 4) since small bandwidth favors bias over variance, we may incorrectly assume there is little harm in using a variable estimate if it is not badly biased.

Unfortunately, severe consequences can result from choosing too large or too small a bandwidth (see supplement Section IX). Most importantly, no systematic method currently exists for bandwidth selection. We propose such a method in this letter.

IV. BIAS OF MULTITAPER ESTIMATES

Multitaper estimates are asymptotically unbiased and are distributed according to a scaled χ_{2K}^2 distribution. The bias of a multitaper spectrum estimator can be written as

$$\mathbf{E}\{\bar{S}(f)\} - S(f) = B_B(f) + B_W(f) \quad (3)$$

where $B_B(f)$ denotes *broadband* expectation, or leakage, and $B_W(f)$ denotes the *local* bias, namely

$$B_B(f) = \frac{1}{K} \int_{\mathcal{D}} \sum_{k=0}^{K-1} |U^{(k)}(f - \nu)|^2 S(\nu) d\nu \quad (4)$$

$$B_W(f) = \frac{1}{K} \int_{-W}^W \sum_{k=0}^{K-1} |U^{(k)}(f - \nu)|^2 S(\nu) d\nu - S(f). \quad (5)$$

where the set $\mathcal{D} = [-1/2, 1/2] \setminus (-W, W)$.

Broadband expectation is bounded above

$$B_B(f) \leq \frac{1}{K} \sum_{k=0}^{K-1} (1 - \lambda_k) \sigma^2 \quad (6)$$

by using the Cauchy inequality [1, Sec. IV, eq. (4.7)], where σ^2 is the variance of the process $x(t)$. The quantity $1 - \lambda_k$ is very small for $k = 0, \dots, K-1$, (see [1, Table II]) so the overall leakage is small compared with the local bias.

Replacing the sum in (5), called the *spectral window*, by the simple rectangular approximation

$$\frac{1}{K} \sum_{k=0}^{K-1} |U^{(k)}(f)|^2 \approx \begin{cases} \frac{1}{2W}, & f \in (-W, W) \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

and assuming the spectrum is twice continuously differentiable, a second-order Taylor series approximation for the spectrum, valid where $\nu \in (f - W, f + W)$, can be employed

$$S(\nu) \approx S(f) + (\nu - f)S'(f) + \frac{(\nu - f)^2}{2!} S''(f). \quad (8)$$

The dominant local bias expression (5) can, then, be approximated as follows, [9], [10]:

$$B_W(f) \approx \frac{1}{2W} \int_{f-W}^{f+W} S(\nu) d\nu - S(f) \quad (9)$$

$$\approx \frac{W^2}{6} \cdot S''(f), \quad (10)$$

where all odd-order derivative terms have disappeared.

A. Bias of the Log Spectrum

The MSE of the log spectrum can be related to the error for the cepstrum [15], defined as the inverse Fourier transform of the log spectrum. Proceeding in the spirit of [12, eqs. (15) and (18)], we see that

$$\Lambda(f; N, W) = \mathbf{E}\{(\log \bar{S}(f) - \log S(f))^2\} \quad (11)$$

$$\approx \left[\log \frac{\mathbf{E}\{\bar{S}(f)\}}{S(f)} \right]^2 + \mathbf{Var}\{\log \bar{S}(f)\} \quad (12)$$

$$\approx \left[\frac{\mathbf{E}\{\bar{S}(f)\} - S(f)}{\mathbf{E}\{\bar{S}(f)\}} \right]^2 + \frac{\mathbf{Var}\{\bar{S}(f)\}}{\mathbf{E}\{\bar{S}(f)\}^2} \quad (13)$$

where the approximation (13) is appropriate for any consistent estimator of $S(f)$, since it ensures that $\mathbf{E}\{\bar{S}(f)\} \approx S(f)$. That is, the MSE of the log-spectrum estimator can be written in terms of the MSE of the spectrum estimator and its expectation.

B. Estimation of Multitaper Local Bias of the Log Spectrum

When the spectrum is twice continuously differentiable, one estimates local bias by estimating the second derivative of the spectrum, as in (10). Local second-derivative estimates of the spectrum have been computed by using Chebyshev polynomials as basis functions as in [9]; however, here we apply an alternative “global” method using splines.

First, one selects a set of M frequencies of interest, say, $\{f_m\}_{m=0}^M$, with $f_m = mW/P$ for some $P > 0$ and $M = \lfloor P/W \rfloor$, then computes an estimate of the spectrum at each of the nodes $\tilde{S}(f_m)$, $m = -M/2, \dots, M/2 - 1$ and fits a spline, defined piecewise over $[-1/2, 1/2)$. Choosing $P = 1/2$ results in $2W$ frequency separation between the samples and ensures approximate independence of the $\tilde{S}(f_m)$ values. The spline tuning parameter, $\approx \sqrt{K}$, is selected as the reciprocal of the standard deviation of the estimator [16] and determines the number of knots. One differentiates the spline twice to obtain an estimate of the second derivative $\tilde{S}''(f)$ of the spectrum [16]. Since the estimate of the spectrum is asymptotically unbiased and consistent, the limiting behavior of the spline estimate of the second derivative is convergence to its true value.

The result is the following expression for the estimated squared bias of the log spectrum (first term of (13))

$$\left[\frac{\frac{W^2}{6} \tilde{S}''(f_m)}{\tilde{S}(f_m)} \right]^2 \quad (14)$$

evaluated at the spline frequencies.

V. VARIANCE OF MULTITAPER SPECTRUM ESTIMATORS

The variance expression for the multitaper is quartic in $x(t)$ and can be found in [1, Sec. VII] for the simple case where the data $x(t)$ are Gaussian. The multitaper approach gives consistent estimates of the spectrum provided that the number of tapers, K , increases with the number of samples, N . The variance can be simply estimated as follows.

A. Jackknife Variance Estimation for Multitaper Spectra

The spectra, $\hat{S}^{(k)}(f)$, produced as intermediates of the multitaper computation, are approximately independent and, hence, can be used to produce an estimate of variance. The *delete one jackknife* method [11], [17], [18], gives an estimate having less bias than one computed directly as a sample variance of the individual eigenspectra [19] and is statistically efficient for computation on a small number of independent samples. Here, we jackknife the logarithms of the spectra, in lieu of the spectrum itself, in order to stabilize the variance of the resulting estimate [20, Sec. 5.3]. The details of this computation are reviewed in the supplement. Denote by $\hat{\sigma}_J^2(f)$, the estimate of the variance of the log spectrum at frequency f . The result roughly follows a t -distribution, so one can use the variance estimate to construct $100(1 - \alpha_0)\%$ confidence intervals for the log spectrum as in [11, eq. (2.49)]. Note that since the multitaper method produces approximately independent estimates, $\hat{S}^{(k)}(f)$, it is the only method for spectrum estimation that is accompanied by confidence intervals [11].

VI. OPTIMAL BANDWIDTH ESTIMATE

When the true log spectrum is unknown, the MSE is estimated by squaring the spline estimate of the bias in (14) and adding the jackknife estimate of the variance $\hat{\sigma}_J^2(f)$ of the log spectrum

$$\hat{\Lambda}(f_m; N, W) = \frac{W^4}{36} \left[\frac{\tilde{S}''(f_m)}{\tilde{S}(f_m)} \right]^2 + \hat{\sigma}_J^2(f_m). \quad (15)$$

Then, by summing over all frequencies to get an overall estimate of the error, the optimal time-bandwidth product can be written as

$$\arg \min_{NW} \sum_{m=-M/2}^{M/2-1} \hat{\Lambda}(f_m; N, W) \quad (16)$$

where N is assumed fixed. By computing (16) for various choices of NW , usually restricting oneself to half-integer values, one can select the bandwidth that produces the estimate with the least estimated MSE.

A. Summary of bandwidth selection

In summary, for each NW , one computes an integrated MSE estimate as follows:

- 1) compute the multitaper estimate of the spectrum (1);
- 2) assuming a second-order Taylor's series is appropriate, estimate the local bias as described in Section IV-B;
 - a) fit a spline to $\tilde{S}(f)$ on a grid of frequencies $\{f_m\}$ spaced $2W$ apart;
 - b) evaluate the second derivative of the spline estimate at these frequencies, denoted $\tilde{S}''(f_m)$;
 - c) estimate $\widehat{B}_W(f)$ as in (10) as the sum $\frac{W^2}{6} \tilde{S}''(f_m)$;
 - d) estimate the squared bias of the log spectrum using (14);
- 3) compute the jackknife estimate of the variance of the log spectrum, $\hat{\sigma}_J^2(f)$, from Section V-A;
- 4) substitute the quantities from 2d) and 3) into (16), and sum over the set of frequencies f_m .

Then, select NW such that the quantity from 4) is minimized.

VII. LORENTZIAN OPTIMAL BANDWIDTH

As an example, we introduce the *Lorentzian* spectrum which models damped oscillatory random processes

$$L(f) = \frac{A\alpha^2}{\alpha^2 + (f - f_0)^2} \quad (17)$$

where f_0 is the frequency (in Hz) of the undamped oscillator, the *half-width at half-maximum* α of the oscillator represents the rate of exponential decay, or damping; and A is the gain. In the simulations that follow, parameters $A = 100$, $\alpha = 0.01$, and $f_0 = 0.2$ have been used.

The relationship between time-bandwidth product and MSE is shown in Fig. 1 by simulating the Lorentzian process with sample size $N = 1000$ 25 times. If one computes as the integrated squared distance between the spectrum estimate (1) and true spectrum (17), one gets the solid curve in Fig. 1. Using our method to estimate (16) one obtains the dashed curve, the result (marked with “x”) is the mean of the bandwidth obtained over the 25 simulated processes. Splines were estimated using a grid of frequencies placed $W/2$ Hz apart, allowing slight correla-

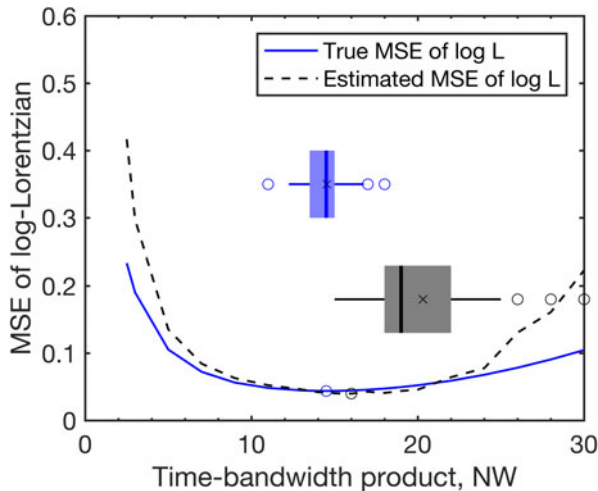


Fig. 1. MSE estimates for the log-Lorentzian (16) versus NW for $N = 1000$ averaged across 25 realizations and all frequencies. Two quantities plotted are the integrated MSE of the log-Lorentzian, obtained by simulating the process 25 times, and computing the squared distance between multitaper spectrum and true value (blue, solid), and its estimated value (black, dashed line). Boxplots show values of the time-bandwidth corresponding to the smallest MSE obtained by simulation (upper boxplot, corresponds to solid blue curve) using the true spectrum, and by estimating the MSE using our method (lower boxplot, corresponds to dashed curve). Open blue circles on the curves represent minima of the means of the MSE curves (of 25 samples). While mean values are in near-agreement (marked with “x”) at $NW = 16$, estimated and $NW = 14.5$, simulated, the estimation procedure tends to give minima of the MSE at a slightly larger time-bandwidth product.

tion between samples, while ensuring dense enough sampling to capture overall behavior. Boxplots show the robust summary statistics for the optimal NW values obtained over the 25 simulations. Note that the minima of these two mean curves are close: the minimum of the estimated MSE occurs at $NW = 16$ on average, while the minimum MSE obtained by simulation occurs near $NW = 14.5$.

The optimal bandwidth choice for the Lorentzian spectrum depends only on sample size and α and is derived in the supplement. Note that this is for the case where the logarithm of the spectrum is not taken. In summary, the bandwidth that minimizes the integrated MSE for the Lorentzian spectrum is asymptotically

$$W_{\text{opt}} = c_0 \left(\frac{\alpha^4}{N} \right)^{1/5} \quad (18)$$

where c_0 is some positive constant.

Fig. 2 shows the optimal NW values for the Lorentzian and log-Lorentzian as N increases. Lower curves represent the optimal NW based on minimizing the MSE computed on the spectrum itself and not its logarithm (18) (not obtained experimentally, but compare with (18)), and solid upper curves represent optimal NW values obtained by minimizing the MSE of the log spectrum. Solid lines represent the difference between the spectrum estimate and its true value, as in Fig. 1. If the MSE of the log spectrum is minimized, one arrives at a larger optimal time-bandwidth product than if one were to minimize the MSE of the spectrum itself.

We conclude that, for this example, the method we described for computing the MSE from the data yields approximately the same curve (on average) that one can obtain by integrating the

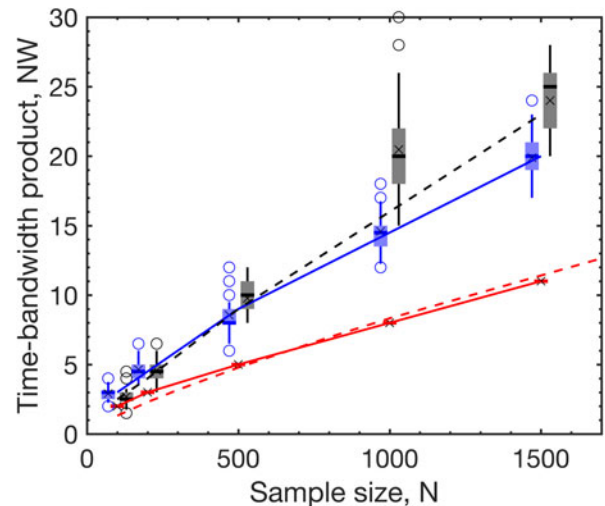


Fig. 2. Optimal time-bandwidth product for the Lorentzian (lower pair of curves) and log-Lorentzian (upper solid curves and boxplots) as a function of sample size. Here, the same parameters are used as in Fig. 1, and mean values over 25 simulations of the process for the optimal time-bandwidth are shown. When the logarithm is not taken, an analytic expression for the optimal bandwidth is derived in the supplement, and is shown as the dashed lower curve, scaled to approximately match the solid lower curve.

squared difference between the log spectrum estimate and its true value; however, as Figs. 1 and 2 show, one can easily obtain large values for time-bandwidth parameter, since large NW can produce spuriously small values for estimated bias.

VIII. DISCUSSION AND CONCLUSION

We have described a technique for estimating the MSE of a multitaper spectrum estimate in terms of its bandwidth. This procedure greatly reduces the subjectivity in the choice of bandwidth for the multitaper method. As shown in the supplement, inappropriate choice of bandwidth has severe consequences; and in none of these examples is a time-bandwidth product of 4 optimal. Fortunately, our method can be implemented automatically so the user will not have to manually choose the bandwidth. We caution that this method is appropriate on regions in frequency where the true spectrum is sufficiently smooth. When conditions are violated, often from discontinuities resulting from line components, one can identify these by using a pilot spectrum estimate and F-test and replace them in a “reshaped” version of the spectrum [1, Sec. XIII-B]. By virtue of being squared magnitudes of linear combinations of complex exponentials, we remind that all spectrum estimates are infinitely continuously differentiable. As such, no spectrum estimator will be able to faithfully reproduce cusps or large jump discontinuities except in the limit of large sample sizes. As is seen in the supplement, the multitaper method performs well as compared to other non-parametric estimators when the true spectrum has large range, and in this respect is ideally suited to estimating the size of large discontinuities.

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